Mohammed Rawih

Senior Software Engineer – Full Stack Engineer

Riyadh, Saudi Arabia | mohammedraweih@gmail.com | +966509556183
linkedin.com/in/mohammed-rawih-48a8a6322

PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer with over 7 years of experience specializing in system analysis, architecture design, and building scalable, production-ready systems using microservices-based architectures. Highly experienced in designing and developing backend platforms with .NET and Python, modern web applications using React and Next.js, and delivering mobile applications as part of fully integrated solutions. Strong background in microservices, API design, and distributed systems, with the ability to decompose complex business domains into maintainable, independently deployable services. Proven ability to own the full software lifecycle — from requirements analysis and system design to development, deployment, and long-term production operation. Known for technical depth, reliability, and building systems that perform under real-world load while remaining secure, scalable, and easy to evolve.

SKILLS

Core Technologies (Production Experience)

- .NET 8/9 (C#) – advanced OOP, async/await, memory management
- ASP.NET Core Web API & MVC (REST APIs, Razor Views, filters, middleware)
- Entity Framework Core (code-first, migrations, performance tuning, tracking/no-tracking)
- Minimal APIs for high-performance microservices
- Background services, hosted services, and Windows Services
- Desktop applications using WPF and WinForms
- Dependency Injection, configuration, logging, and caching
- Authentication & Authorization (JWT, OAuth2, Identity, RBAC)
- Python (FastAPI, Django)
- C++ (17+) for performance-critical components
- React.js & Next.js 15 (App Router, Server Components, SSR)
- Angular (enterprise web applications)
- Mobile application development with Flutter (iOS & Android)
- Mobile-backend integration, offline mode handling, performance optimization, and secure storage

Data Platforms & Storage

- PostgreSQL (enterprise-grade systems, heavy indexing, partitioning, and performance tuning)
- SQL Server (large-scale ERP systems, stored procedures, replication)
- MySQL / MariaDB (high-traffic systems with sharding experience)
- Redis (distributed caching, pub/sub, session and state management)
- Supabase (authentication, realtime, and storage in production)

System Architecture & Distributed Systems

- Clean Architecture and Vertical Slice design
- Domain-Driven Design (DDD), CQRS, MediatR

- Microservices architecture and service decomposition
- Event-driven systems and asynchronous messaging (RabbitMQ)
- Distributed systems design and fault-tolerant services
- Advanced data modeling and query optimization for large-scale systems

API Design & Integration

- RESTful API design, versioning, and documentation
- Secure service-to-service communication
- API gateways, webhooks, and third-party integrations
- Idempotency, retries, and resilient API design

DevOps, Cloud & Operations

- Docker and Kubernetes for containerized deployments
- CI/CD pipelines using GitHub Actions
- Production deployments and environment configuration
- Alibaba Cloud and Microsoft Azure (production experience)

Production Support & Reliability

- Production monitoring, logging, and alerting
- Incident handling, root cause analysis (RCA), and hotfix delivery
- Performance troubleshooting in live environments
- High-availability and uptime-focused system design

Security & Quality

- Secure API design using JWT, OAuth2, RBAC, and rate limiting
- Access control and permission management
- Unit and integration testing using xUnit
- Test-driven development (TDD)

Applied Artificial Intelligence

- Integration of AI capabilities into backend systems
- Experience with Large Language Models (LLMs) and prompt design
- Retrieval-Augmented Generation (RAG) pipelines
- Computer vision and image processing workflows
- Classical machine learning algorithms (e.g., KNN)

WORK EXPERIENCE

Full Stack Developer

Maham IT Solutions, Riyadh, Saudi Arabia **Feb 2024 – Present**

- Developed key modules of the Maham-AI ERP system (HR, CRM, Services) and integrated AI-driven analytics to enhance automation and decision-making.
- Designed and implemented PostgreSQL relational databases, ensuring scalability, data integrity, and optimized query performance.
- Implemented full-stack features using React and .NET Web API, fully integrated with Supabase services for database, authentication, and real-time synchronization.
- Improved system performance through SQL optimization, Redis caching, and backend refactoring, reducing response time and server load.
- Implemented a complete Role-Based Access Control (RBAC) system, enabling secure permissions and multi-level user access.

- Managed GitHub repositories, enforced code standards, and maintained CI/CD pipelines for continuous delivery.
- Deployed and monitored production environments on Alibaba Cloud, ensuring high availability, security, and reliable performance.

Backend Developer

HowbaniSoft Riyadh, Saudi Arabia(remotely) **Dec 2020 – Jun 2023**

- Designed, developed, and maintained backend systems using .NET ,Java technologies.
- Built, optimized, and managed SQL Server databases to improve performance and reliability.
- Developed secure and scalable RESTful APIs for various business applications.
- Implemented Docker to containerize services and streamline deployment workflows.
- Utilized GitHub for version control, code reviews, and collaborative development.
- Contributed to system architecture improvements and backend performance optimization.
- Worked efficiently under pressure, delivering high-quality backend solutions that enhanced the company's technical operations.

Full-Stack web and Mobile Developer

Al-Rahal for IT Solutions And Web Services, Riyadh, Saudi Arabia (Freelance) **Jan 2018 – May 2019**

- Built and deployed user-friendly and responsive web applications using HTML, CSS, JavaScript, and React.js.
- Designed and implemented scalable and secure back-end systems utilizing .NET, and Django frameworks.
- Created robust Fast APIs to facilitate efficient data exchange between front-end and back-end applications.
- Led code reviews to ensure code quality and adherence to coding standards and best practices.
- Collaborated closely with cross-functional teams (e.g., marketing and designers) to analyze requirements, design solutions, and ensure successful project implementation.

Key Projects

OneMay – Enterprise ERP & Operations Platform (Boulevard & Riyadh Seasons)

- Led the development of a large-scale, **enterprise-grade ERP and operations platform** designed to support **Boulevard management and Riyadh Seasons–scale events**.
- Delivered a unified system combining **core ERP capabilities** with advanced **event and crowd operations**, built to operate in high-traffic, mission-critical environments.
- Implemented comprehensive ERP modules covering **Finance, Human Resources, Customers, Vendors, Contracts, Procurement, and Operational Reporting**.
- Designed and developed specialized systems for **event management, gate and access control, crowd management, risk management, and operational coordination**.
- Built the platform using **Next.js, React, and TypeScript** on the frontend, with **.NET 8 / ASP.NET Core** powering backend services.
- Developed and integrated **30+ enterprise modules** supporting both administrative and on-ground operational workflows.
- Designed and implemented **200+ RESTful APIs** and **100+ production-ready screens**, supporting complex, role-driven business processes.
- Established a scalable frontend architecture using **reusable components and a monorepo (Turborepo)**, reducing development time by approximately **70%**.
- Successfully migrated the frontend from **Vue 3 to React / Next.js** while preserving full system functionality and operational stability.

- Implemented critical features including **full Arabic/English bilingual support with RTL**, **QR-based ticketing**, **role-based access control (RBAC)**, and secure payment gateway integrations.
- Optimized data models and performance using **PostgreSQL and SqlSugar**, supported by **Redis caching** to ensure low-latency operation under peak seasonal traffic.
- Designed the system to meet **high availability, performance, and reliability requirements** expected in large-scale government and public-sector environments.

Legatoo AI platform for analyze the legal cases

- Developed an AI-driven legal consulting platform tailored specifically for **Saudi legal regulations and practices**.
- Built a web-based system for **contract analysis and legal case review**, capable of identifying ambiguous, conflicting, or non-compliant clauses.
- Implemented **Retrieval-Augmented Generation (RAG)** pipelines to enable context-aware legal question answering based on Saudi laws and regulations.
- Integrated AI capabilities to support **legal consultation workflows**, providing actionable insights aligned with local legal frameworks.
- Enhanced accuracy and efficiency in legal document validation, significantly reducing manual review effort for legal professionals.
- Designed the platform to function as both a **legal analysis tool** and an **interactive legal inquiry system** for lawyers and consultants.

Maham-AI ERP – AI-Powered Enterprise Resource Planning System

- AI-powered ERP platform designed to automate enterprise operations across HR, CRM, and service/location management. Built using .NET , React, and Supabase with PostgreSQL.
- Developed key operational modules including HR Management System and Boulevard Rental Tracking System for managing shops, leases, and service areas
- Designed and implemented a scalable database architecture (PostgreSQL) ensuring data integrity, modular structure, and optimized relational models
- Built responsive and interactive frontend interfaces using React, fully integrated with Node.js APIs
- Implemented a complete Role-Based Access Control (RBAC) system enabling granular permissions and secure access across all modules
- Enhanced system performance through SQL query optimization, caching strategies, and backend refactoring
- Collaborated within an 8-member team to design, develop, and deliver core ERP functionalities and AI-driven features
- Integrated AI capabilities using OpenAI API for automated insights, data analysis, and workflow assistance
- Leveraged Supabase for authentication, storage, and real-time features, enabling a modern cloud-driven architecture

Ra'ayati Secure, Scalable EHR Platform

- Led the design and development of an enterprise-grade delivery and logistics platform tailored for the Saudi market.
- Built a unified system connecting customers, merchants, and drivers, managing the full delivery lifecycle from order creation to settlement.
- Designed a scalable microservices architecture (12 independent services) following Domain-Driven Design (DDD), Clean Architecture, and CQRS principles.
- Developed core systems for order management, dispatching, real-time delivery tracking, payments, driver management, and merchant operations.
- Implemented event-driven communication using RabbitMQ to enable reliable asynchronous workflows and service decoupling.

- Built backend services using .NET 10 / ASP.NET Core with Entity Framework Core and PostgreSQL (database-per-service pattern).
- Integrated Redis for distributed caching, improving performance and system responsiveness under high load.
- Implemented secure authentication and authorization using JWT, refresh token rotation, and optional 2FA.
- Developed an enterprise admin dashboard using Next.js 14, React 18, and TypeScript with full Arabic/English bilingual support and RTL.
- Designed and implemented 100+ RESTful APIs supporting complex business workflows and role-based access control (RBAC).
- Enabled real-time capabilities for order and fleet tracking using WebSockets.
- Containerized the entire platform using Docker and Docker Compose, enabling independent deployment and CI/CD readiness.
- Designed the system to meet scalability, availability, and reliability requirements for high-traffic delivery operations.

Ra'ayati Secure, Scalable EHR Platform

- Contributed to the development of “Ra'ayati”, a unified national Electronic Health Record (EHR) platform designed to centralize medical records for citizens.
- Participated in designing a highly scalable and secure architecture supporting nationwide healthcare entities.
- Implemented robust user and permission management with strict access control to protect sensitive medical data.
- Supported complex multi-role healthcare workflows, including doctors, nurses, pharmacy staff, and administrators.
- Enabled interoperability between public and private healthcare providers to improve treatment accuracy and continuity of care.
- Supported full electronic healthcare operations, including diagnoses, prescriptions, and longitudinal medical history management.

EDUCATION

Bachelor of **Engineering in Telecommunications and Information Technology**
University of Taiz , GPA: 83.35% (Ranked 5th in Class) **Graduated: 2017**

CERTIFICATIONS

- Web Application Design and Development, Global Human Development and Research Center.
- Field Training, SabaFon, Telecom, Nov 2023 – Dec 2023
- CCNA Routing and Switching Essentials, 1 Apr 2019

LANGUAGES

- Arabic (Native)
- English (Proficient)

ADDITIONAL INFO

- Iqama (Valid for 2 Years)